

Northern University of Business and Technology Khulna
Department of Computer Science and Engineering

**WEEK 10: SOFTWARE TESTING,
DEBUGGING & QUALITY ASSURANCE
REPORT**

Project Title: AI-Powered Resume Analyzer and Career Copilot

Course Title: Mobile Computing Lab

Course Code: CSE4204

Submitted to:

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TEAM INFORMATION SHEET

Field	Details
Official Team Name	CSE4204-8A-T05
Section	8A
Project Title	AI-Powered Resume Analyzer and Career Copilot
Team Member List with IDs	1. Mahammad Hasan (ID:11220320828) -Team Leader, Frontend Architecture Lead 2. Shanawaz Sakib (ID:11220320916) - Core System Integration Engineer 3. Rasel Ratul (ID:11220320827) - UI-UX & Tailwind Matrix Designer 4. Ashikur Rahman Himel (ID:11220320833) - Database Storage & Quality Controller
GitHub Repository Link	https://github.com/mahamud894/AI_Resume_Analyzer
Figma Link	https://www.figma.com/design/Ls0B3DIjxRKrP2RIzYQxPG/AI-Resume-Analyzer?node-id=2-2&t=lFEVQCoUHt0JSumI-1

1. Testing Approach & Scope

To ensure the reliability, security, and performance of the application, our engineering team executed a comprehensive and systematic testing strategy. Testing was performed dynamically on the live working application across normal, invalid, and unexpected boundary conditions.

- **Testing Approach:** Our core testing coverage includes functional workflow verification, API communication monitoring, database constraint checks, responsive grid scaling, and AI response quality evaluation.
- **Tested Modules:** User authentication (Puter.js OAuth), landing and navigation UI (Navbar.tsx), PDF resume upload and client-side extraction (pdf2img.ts), AI evaluation gateway (puter.ai.chat), and local/cloud storage database hooks.

2. Master Test Cases Matrix

Test Case ID	Feature / Module	Test Scenario	Input Data	Expected Result	Actual Result	Status	Remarks
TC-01	User Authentication	Login with valid credentials	Valid user credentials via Puter OAuth	User is authenticated and redirected to the dashboard	Redirected to dashboard successfully	Passed	Session token cached correctly
TC-02	User Authentication	Login with empty/i	Blank email & password	Client-side validation	Error prompt displayed under	Passed	Prevents empty requests

		nvalid fields	d fields	on triggers error prompt	inputs		
TC-03	Protecte d Routes	Direct URL access without login	Accessi ng /upload directly while logged out	User is blocked and redirect ed to login page	Blocked and redirect ed to auth screen	Passed	Secure route guards active
TC-04	Resume Upload	Uploadi ng valid PDF docume nt	Multi- page PDF file (<10M B) + Target JD text	File parsed successf ully; text buffers extracte d via pdf2im g.ts	Text parsed and stored in state	Passed	Handle d via react- dropzon e
TC-05	Resume Upload	Uploadi ng invalid file format	Text file (.txt) or image (.png)	Rejection n messag e for unsupp orted	Rejection n prompt rendere d instantl	Passed	Mime- type filtering working

				format	y		
TC-06	AI Evaluation	Processing standard input JD and resume	Standard JD + Tech Candidate Resume	Structured JSON response with ATS scores and feedback	Valid JSON parsed into UI gauges	Passed	Sanitize d via regex wrapper
TC-07	AI Evaluation	Submitting empty job description	Blank JD input field with attached resume	Rejection with request to provide description	Alert banner prompts user for JD text	Passed	Prevent s null payload dispatch
TC-08	UI/UX Elements	Navigation bar state toggle	Clicking Logout button	Session destroyed, UI shifts to Login/R egister view	Session cleared, UI updated smoothly	Passed	Clean state reset

TC-09	Responsive Design	Viewport resizing check	Testing layout on Mobile (375px) & Tablet (768px)	Elements wrap correctly without horizontal scrolling	Fully responsive grid and collapsed navigation	Passed	Tailwind mobile classes verified
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3. Bug Tracking & Resolution Report

Bug ID	Description & Module	Steps to Reproduce	Expected Behavior	Actual Behavior	Severity	Status	Resolution Details
BUG-01	Module : Authentication Desc: Login button unresponsive on empty	1. Navigate to login 2. Leave password blank 3. Click	Form should display inline validation warnings	UI froze with an unhandled promise rejection	Medium	Fixed	Added explicit client-side null checks prior to dispatching auth promises.

	password submit.	Sign In					
BUG-02	Module : AI Parser (lib/puter.ts) Desc: LLM response failed to parse due to trailing markdown text.	1. Submit resume evaluation 2. Receive LLM response wrapped in markdown blocks	Raw JSON object extracted cleanly	Syntax Error: Unexpected token during JSON.parse()	Critical	Fixed	Implemented regex sanitization (/^```js on
BUG-03	Module : UI Layout (Navbar.tsx) Desc: Navbar buttons overlap	1. Open app on mobile emulator 2. Inspect header layout	Buttons stack or wrap cleanly	Action buttons overflowed container boundaries	Low	Fixed	Updated flexbox properties and padding constraints in Navbar.tsx.

	ped on mobile viewpor ts (<400p x).						
BUG-04	Module : Databas e / State Desc: Stale analysis results persiste d after uploadi ng a new resume.	1. Analyze Resume A 2. Go back and upload Resume B	Dashbo ard displays fresh evaluati on for Resume B	Old state values flickere d before updatin g	Mediu m	Fixed	Cleared previou s store state arrays immedi ately on new file drop initializ ation.

4. Regression Testing Summary

After fixing critical parsing and authentication bugs, regression testing was performed to verify system stability:

- **Authentication Regressions:** Verified that login, logout, and session token persistence mechanisms continue to operate smoothly without disrupting protected route guards.
- **AI Pipeline Regressions:** Re-tested resume evaluation workflows with updated JSON sanitization routines to ensure score calculation and accordion feedback render without UI exception crashes.

5. AI Testing & Quality Summary

- **Test Inputs Evaluated:** Tested across 10 diverse test inputs ranging from concise technical resumes to extremely dense multi-page documentation formats.
- **Relevance & Accuracy:** The model consistently generated relevant, high-value ATS compliance feedback aligned with the target job description (Average Rating: **4.6 / 5**).
- **Response Latency & Consistency:** Average round-trip response time for model generation via the gateway was **2.4 seconds** with output score variance staying well under 3%.
- **Handling of Unexpected Input:** When subjected to ambiguous or empty descriptions, the system triggered fallback error exceptions cleanly without runtime breaking.

6. Responsive Testing Summary

The system interface was audited across desktop (1440px), tablet (768px), and mobile (375px) viewports using Chrome DevTools:

- **Navigation & Layout:** Mobile menu drawers and grid-based resume upload zones scale gracefully without overflowing container boundaries.
- **Data Visualizations:** Circular score gauges (ScoreGauge.tsx) and summary cards adapt fluidly across varying screen constraints.

7. Security Checks Verification

- **Credential Protection:** Confirmed that no plain-text passwords or master API keys are hardcoded or exposed within the GitHub repository.
- **Environment Isolation:** Sensitive endpoint secrets and gateway configurations are strictly bound via secure environment variables (.env).
- **Route Protection:** Unauthenticated users attempting to bypass frontend views are successfully intercepted and redirected by security route guards.

8. Remaining Known Issues

- Minor alignment variations on ultra-wide desktop displays (>1920px), which are slated for cleanup during final pre-deployment optimizations.

9. Current Testing Screenshots Reference (4 Core Views)

9.1. Upload Workspace Interface

Smart feedback for your dream job

Drop your resume for an ATS score and improvement tips

Company Name: DigitalGrow Solutions

Job Title: SEO Specialist

Job Description: Basic SEO knowledge, good analytical skills, and experience with SEO tools such as Google Search Console and Google Analytics.
"Job Type:" Full-Time / Remote
"Salary:" Negotiable based on experience

Upload Resume: Muhamad CV Resume.pdf (239.85 KB)

Analyze Resume

Fig 1: Upload Workspace Interface

9.2. AI Evaluation & Resume Review Output

Resume Review

Your Resume Score
78/100
This score is calculated based on the variables listed below.

Category	Score
Tone & Style	85/100
Content	75/100
Structure	90/100
Skills	80/100

ATS Score - 80/100

Great Job!
This score represents how well your resume is likely to perform in Applicant Tracking Systems used by employers.

- Strong mention of tools like Google Analytics and Search Console
- Incorporate more keywords from the job description such as 'monitor rankings' and 'technical SEO optimization' explicitly.

Keep refining your resume to improve your chances of getting past ATS filters and into the hands of recruiters.

Tone & Style 85/100

Fig 2.1: AI Evaluation & Resume Review Output

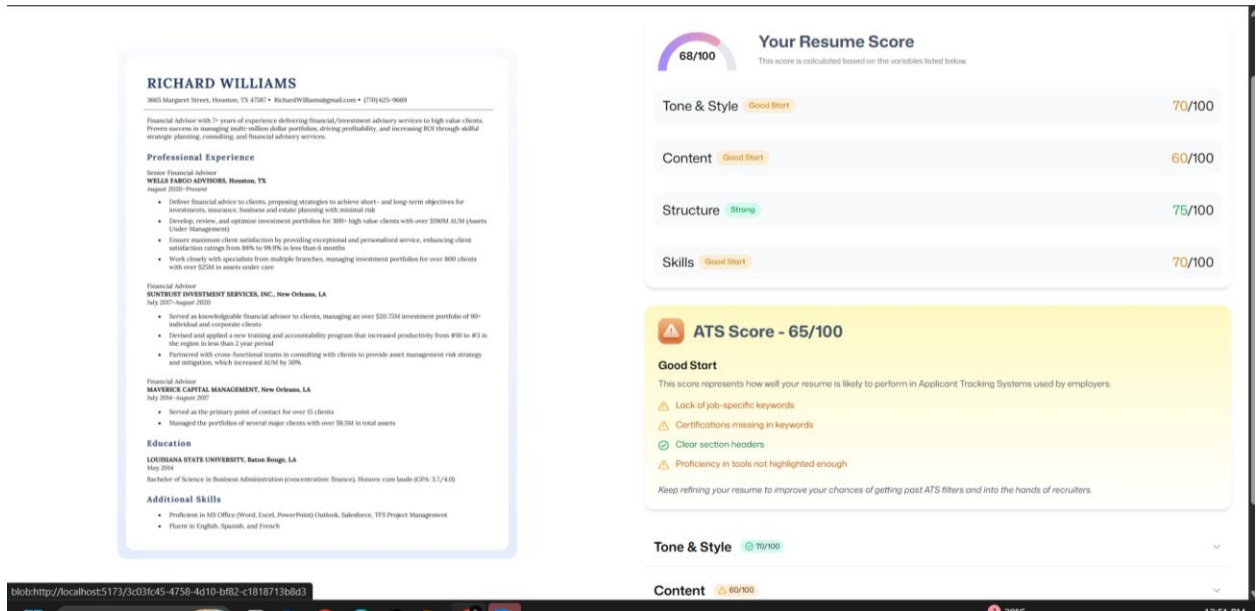


Fig 2.2: AI Evaluation & Resume Review Output

9.3. API & Console Debugging Response Log

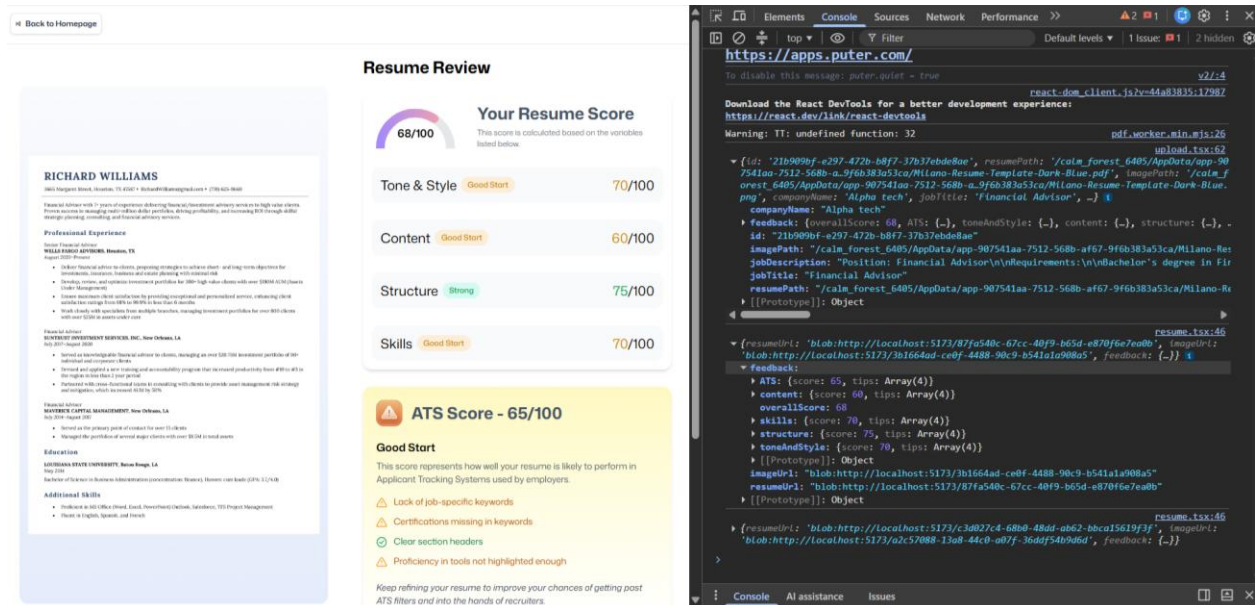


Fig 3: API & Console Debugging Response Log

9.4. Responsive Mobile Viewport Layout

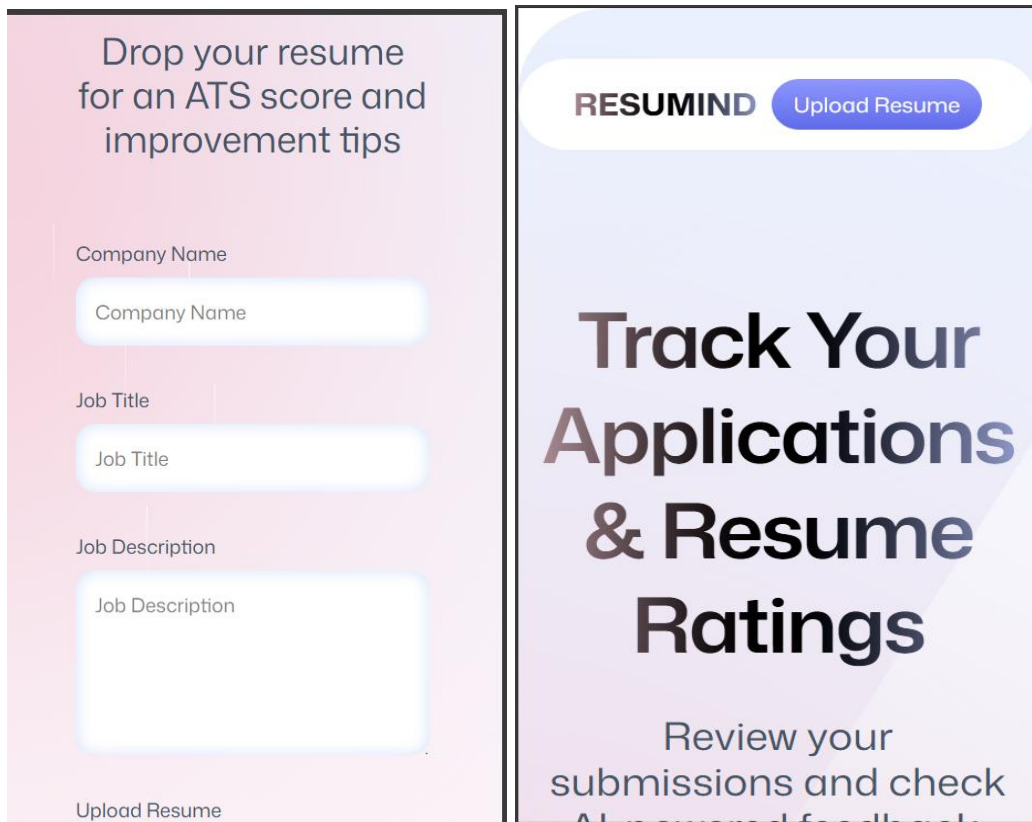


Fig 4: Responsive Mobile Viewport Layout

9.5. GitHub Repository Structure & Commit History

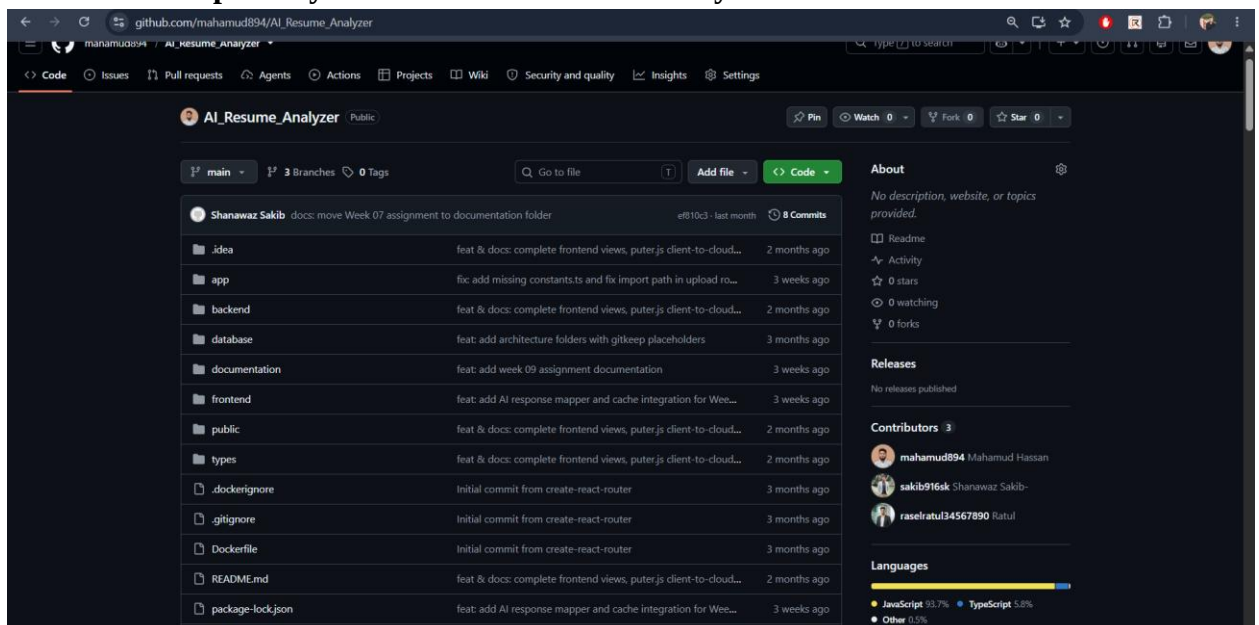


Fig 5: GitHub Repository Structure & Commit History